

Knowledge Management in a World of Generative AI: Impact and Implications

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The management of organizational and worker knowledge is a critical part of any organization. Knowledge management recognizes the explicit knowledge that workers contribute to an organization, as well as the organizational knowledge and skills acquired over time. The potential impact of generative AI (GenAI) raises many issues and opportunities related to effective knowledge management as organizations explore the impact and implications of its adoption. This research explores issues related to an increasing world of GenAI and proposes a *Framework for Knowledge Management in the GenAI Era*. Knowledge management workers, who are novices, might be more inclined to use generative AI for performing their tasks. Expert knowledge workers might focus on quality evaluation and quality assessment. Adoption of GenAI may help with the progression from novice to expert, while freeing up time and resources for other work. Challenges include the potential loss of critical organizational knowledge, definition of job-related requirements, expectations, and tasks, as well as over-reliance and loss of tacit or unarticulated knowledge. These challenges must be balanced against available access to large amounts of general and domain-specific knowledge from GenAI to capture the value realized from its adoption.

CCS Concepts: • **Information systems** → **Information systems applications**; **Decision support systems**;

Additional Key Words and Phrases: Knowledge management, generative AI (GenAI), novice knowledge workers, expert knowledge workers, Framework for Knowledge Management in the GenAI Era, risk assessment, organizational knowledge, Knowledge shifting

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1 Introduction

Large language models (LLMs) are deep learning models trained on very large sets of data and can perform reasonably well at language processing tasks. They generate new text by predicting the likelihood of word sequences, based on the given input [8, p.2]. **Generative artificial intelligence (GenAI)** is built on LLMs and algorithms that have led to the emergence of ChatGPT and other AI tools. These tools are creating much excitement and enthusiasm, as well as skepticism and concerns about the potential impact of GenAI, and labeling it as a disruptive technology.

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GenAI systems are based on empirical and probability-based relationships between words and use neural networks to create dialogues with the user. GenAI systems are AI systems that can produce content such as text, images, music, programming code and other outputs.¹ Earlier work on GenAI included **generative adversarial networks (GAN)** [15]. More recent work is based on the ability of large language models that use a transformative ability to predict word sequences to generate new text [8]. However, as noted by Saba [34, p.3], "... due to their sub-symbolic nature, whatever 'knowledge' these models acquire about how language works will always be buried in billions of microfeatures...none of which humans can meaningfully interpret. This is also the reason why explainability can never be attained in such models..."

Interest in GenAI has been motivated by the use of tools such as ChatGPT, MidJourney, Bard (Google), and CoPilot AI. ChatGPT, probably the most popular GenAI tool, processes large amounts of data to create new content (essays, computer code, lyrics). LLMs make it possible for AI systems to assess human context in relatable and plausible ways [7], allowing these systems to perform knowledge-centric tasks they could not accomplish previously. Such capabilities are considered novel, with the potential to support many business cases and, therefore, should be useful for knowledge management. Analyses of ChatGPT, especially, has concluded that GenAI can be used for many activities, resulting in new and efficient ways to perform tasks, albeit with concerns of large-scale plagiarism and misinformation [37].

GenAI can be used to create new combinations of text, with the notion behind large language models being that they can be predictive and generative and able to respond to queries to which only humans have been able to respond in the past. Large language models are becoming platforms for applications that can help provide information to users to support their decision making [28]. OpenAI's evaluations of latest versions of ChatGPT show that it performs exceptionally well on a variety of tests and that it can "reason" better than prior versions and address harder problems, for example, in science, math, and healthcare, as it continuously evolves.² This claim is due to training the systems to try different strategies and recognize mistakes. Huang and Chang [17] provide a survey of reasoning in LLMs and conclude that there has been a great deal of progress in natural language, but the extent to which LLMs are capable of reasoning, as opposed to memorizing patterns, for example, remains unclear. The researchers also provide a classification of various types of reasoning that they attempt to apply to LLMs.

The capabilities of humans to gain and use knowledge has long been recognized in the area of knowledge management [25]. Knowledge management refers to the ability of a firm to develop, share, and leverage their organizational knowledge and is considered critical for organizations to innovate and compete [3]. Knowledge work has traditionally consisted of well-defined and well-structured tasks, as well as tasks that require creativity and insights. The importance of managing knowledge has existed for decades. Now, **artificial intelligence (AI)** is considered a revolutionary technology that could have profound implications for businesses and society on knowledge acquisition and use. Much of this view has cumulated in the release of ChatGPT by OpenAI, which was quickly adopted as users began to explore its usefulness and effectiveness [11]. Since knowledge management deals directly with organizational knowledge, it is reasonable to assume that knowledge management and knowledge workers could be affected deeply by GenAI. The challenges are to identify how GenAI can be used and what value might be derived from such use.

¹Artificial intelligence (AI) refers to "the ability of computers to perform tasks that have historically required human cognition and other intellectual abilities, such as perception, abstraction, inference, learning, and decision making" S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*. Pearson, 2016.

²<https://www.cnn.com/2024/09/13/tech/chatgpt-openai-o1-human-reasoning/index.html>

The promise of GenAI is to support organizations in the automation and support of tasks previously conducted solely by humans. The characteristics of tasks that would benefit the most from GenAI are not well-identified; nor are they well-defined. With today's emphasis on the importance of GenAI, this leads organizations to consider how to create an adoption strategy for its use, even though its potential value might not be known in advance or might be difficult to realize. At the same time, research is fundamentally needed on GenAI as a new and evolving technology. This topic has implications for organizational impact and applications while acknowledging technology challenges.

The objective of this research is to examine the potential impact of GenAI from the perspective of knowledge management and the knowledge workers who might be significantly impacted by this technology. Specifically, this research examines the types of knowledge workers who might be affected by GenAI and proposes a *Framework for Knowledge Management in the GenAI Era*.³ The framework highlights, and possibly revitalizes, the importance of traditional knowledge management. The contributions are to: identify the potential importance of GenAI end products for knowledge management; provide a framework for understanding the impact and implications for organizations; and specify a set of issues that could pose potential future challenges and topics that will require careful consideration and research.

The next section reviews related research on GenAI and knowledge management and its role within organizations. Section 3 proposes the framework for GenAI. Section 4 outlines the issues and implications. Section 5 proposes future research topics, and Section 6 concludes the article.

2 Related Research

Both GenAI and knowledge management deal with some form of knowledge. GenAI is intended to support or replace the knowledge acquisition capabilities of humans by providing access to large amounts of knowledge in a form that can respond to user queries in a manner that has not been previously possible in natural language processing. Knowledge management concentrates on the workers within an organization who acquire and share both explicit and tacit knowledge deemed valuable to an organization. Hence, GenAI should have a significant impact on it. At the same time, decades of research on knowledge management should be useful for understanding GenAI.

2.1 Generative AI

LLMs are deep neural networks, pre-trained using very large amounts of text with millions or even billions of parameters [9]. They form the basis for Generative AI, of which ChatGPT has become widely recognized, including securing recognition as the Economist's 2023 word of the year.

Large Language Models can be considered as bottom-up reverse engineering of language at scale [34]. These language models have valid patterns of linguistic communication by predicting the next sequence of words that a human would normally express based on deep learning and transformer architectures [9]. Because they work on predictions, they are considered "generative." There is much interest in analyzing the potential impact of GenAI within organizations [26]. There is much interest in understanding and analyzing the evolution and impact of these models [5, 9, 14, 41].

Massive language models have "memorized" (put into syntax and represented) many, if not most, valid patterns of linguistic communication [45]. ChatGPT, for example, attempts to create a reasonable continuation of the text it has and might be what is expected to be written after "seeing what people have written on billions of webpages" [45]. This makes for uncertainty associated with the outcome, requiring some type of validation before use, as one would expect

³An era is a recognized stage of development (<https://www.merriam-webster.com/dictionary/era>).

from expert knowledge workers. In this sense, GenAI is not an attempt to understand how the human brain works, but rather, how to build an artificial version of one that does not have a sense of understanding as a human would.⁴ Rather, it has access to a very large database that consists of facts (explicit knowledge) and heuristics (tacit knowledge). The subsymbolic representation in neural networks does not facilitate explanation, because the output, as produced by a combination function that could have many inputs, is not definitive [16, 35].

Probably the people who will be most impacted by this technology are the knowledge workers; those considered critical for innovation and organizational memory [2, 3]. Distinctions can be made between early adopters and those who have already gained the knowledge crucial to contribute to an organization. It might now be possible to speed up tasks or knowledge acquisition that might have taken perhaps a year to accomplish in the past to a few months, or even weeks, thereby gaining knowledge within an organization while being aware of the risks.

2.2 Knowledge Management

A knowledge-based view of a firm asserts that organizations can and should exercise oversight on the knowledge held by employees to facilitate knowledge processes, including the “creation, storage, transfer, and application of organizational knowledge” [2, p.114]. Organizational knowledge refers to the knowledge that resides in the members of the organization as well as organizational policies, procedures, and artifacts. Organizations are the guardians of the knowledge within them and must put in place a system and culture that is oriented toward knowledge discovery, sharing, and use [3]. This has led to systematic efforts to identify, capture, represent, and store knowledge to enable its access (and hence, transfer) to others within the organization, as well as to ensure the appropriate application of knowledge across organizational units. The transfer of knowledge is needed “to avoid reinventing the wheel, to learn from best practices, to know where/in whom knowledge resided in an organization and to appropriate valuable knowledge from one part of the firm to another” [3, p.4].

Knowledge work involves cognitive processing of information by knowledge workers to provide value to an organization. It has been proposed that GenAI can support knowledge workers to improve their productivity by: reducing the time needed to perform tasks; reducing a worker’s cognitive load; and helping workers learn or acquire new knowledge, while moving organizations to the next phase of knowledge management [4].

Knowledge workers can be considered as two types. Novice knowledge workers often work with explicit knowledge, which is objective and transferable. Expert knowledge workers possess valuable implicit, or tacit, knowledge. Tacit knowledge can be highly subjective, consisting of insights and intuition that are hard to capture, represent, and share, because they are mostly inherent to an expert knowledge worker [24]. Then, the challenge is how to ensure that explicit knowledge can be translated to tacit knowledge and contribute to knowledge management by applying the tacit knowledge to new situations or innovations.

Knowledge management is concerned with both content and collaboration by converting data and information to connect people to other people and knowledge [27]. It has been proposed that applications of large language models may have the potential to give knowledge management a new importance after discussion of its loss of importance [28].

There has long been a basic understanding that organizations need to access and manage their own knowledge bases that serve as a corporate asset. Thus, managers should focus on the proper

⁴If You Only Learn One Historical Fact about AI, Let It Be This One: AI’s Greatest Lie and Greatest Success, by Alberto Romero: <https://albertoromgar.medium.com/if-you-only-learn-one-historical-fact-about-ai-let-it-be-this-one-4373e94a5092>

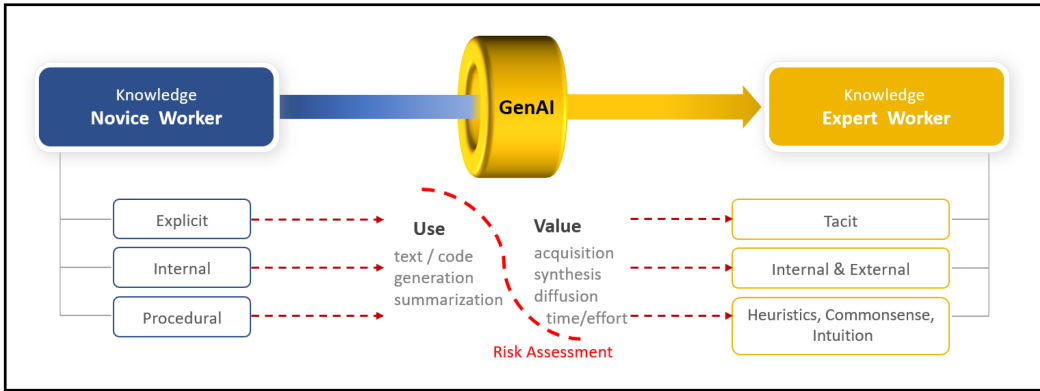


Fig. 1. Framework for Knowledge Management in the GenAI Era.

management of GenAI because of its potential impact on knowledge workers, as well as its perceived disruptive capabilities.

3 Framework for Knowledge Management and Generative AI

Knowledge workers are required to gain knowledge and apply it to activities within an organization that will contribute value to that organization. Companies in all industries rely on knowledge workers to enhance their products, services, and customer support. As knowledge workers acquire and understand the importance of knowledge, they progress from novices to experts.

Figure 1 proposes a framework that represents how GenAI could impact knowledge management. The framework was developed following a conceptual research approach [18]. The research domain is GenAI, with the concept under investigation being its potential application to knowledge management. Multiple concepts and theories are considered (GenAI, knowledge management, organization impact). The concepts are integrated based on: the literature on LLMs and AI; knowledge management and interviews with early adopters and academic experts. The intent of this framework, as is the purpose of all frameworks, is to provide a general, and useful, way to understand a phenomenon [21]; in this case, novice and expert knowledge workers and how they might be impacted by GenAI.

3.1 Knowledge for Novices versus Experts

Novice workers need to gain knowledge that is explicit, internal to an organization, and procedural. As they progress to become even more valuable, expert knowledge workers acquire tacit knowledge and expertise. Their knowledge base expands to gain more of an understanding of external knowledge and how it can impact the internal operations of their company. Explicit knowledge can normally be articulated and expressed in some formal manner [30]. It can be acquired through documentation, study, or other such means. Internal knowledge pertains to how an organization works, its policies and mission, and its positioning within an industry. Procedural knowledge represents the rules, regulations, and procedures a company has in place and follows to carry out its operations.

Expert knowledge workers gain tacit knowledge over time. Tacit, or implicit, knowledge refers to “a kind of knowledge that human beings only develop through the experience gained over years and which cannot be passed on by a set of instructions, manuals and so forth” [30, p.338]. Tacit knowledge is important, even though resources might often be focused on explicit knowledge, which may be easier to manage [39]. It includes “less specifiable insights and skills

that are ‘embedded’ in individuals or an organizational context” [22, p.2]. Since it is often based on expertise, tacit knowledge can involve decision making using heuristics and common-sense knowledge and reasoning.

Organizational knowledge is specific and important to an organization. It is the “capability members of an organization have developed to draw distinctions in the process of carrying out their work, especially concrete contexts, by enacting sets of generalizations whose application depends on historically evolved collective understandings” [43, p.974]. Procedural knowledge refers to the knowledge needed to get things (tasks) accomplished.

3.2 GenAI Use

Although many companies and organizations are finding potentially innovative uses for generative AI, the framework concentrates specifically on knowledge management. This should be advantageous, because the importance of knowledge management has been recognized for decades, with much work on understanding the role of knowledge in organizations and the need for approaches to managing it [2]. Organizations have boundaries [39], so, with the increasing emphasis on a knowledge-driven economy, disseminating knowledge is now considered a high-efficiency organizational behavior [38].

Knowledge acquisition, documentation, sharing, and transfer. Less-experienced knowledge workers can use GenAI to perform tasks faster and more efficiently [3]. GenAI might be able to sort and rank knowledge, although it is always difficult to document implicit or tacit knowledge.

Support for mundane or tasks of minimal importance. GenAI can streamline tasks such as the creation of Excel documents or the writing of some code, including SQL queries. It can generate initial drafts of letters or marketing campaigns based on a standard template. Further support is found for grammar and spell-checking. Employees can make queries about organization-specific information such as organization charts, the history and background of a company, or even the acronyms used, which can be helpful for new employees [46]. Use of GenAI may simplify mundane tasks and free up knowledge workers for other tasks.

Summarization and synthesis. The use of GenAI can focus on certain types of higher-level support activities. This includes summarizing texts or numbers at an aggregate level. It could also include generating timelines such as the progression of an innovation from conception to production or the health history of a victim in a lawsuit.

3.3 Generative AI Value

As with the adoption or adaptation of all new technologies, there is much interest in assessing the value that GenAI can provide to an organization. Many organizations expect that GenAI will transform their organizations within a relatively short, three-year period [11]. Of course, there are many factors to consider when assessing the value of any technology, including the knowledge of workers, efforts for growth and innovation, risk assessment, and societal and regulatory impacts in a global environment [11].

Organizations usually implicitly assume that knowledge can be, and is worth, managing, because they dedicate resources to it [39]. There can be strategic implications; for example, knowledge transfer may contribute to accelerated learning and help nurture clients or innovation. One impact of generative AI may be job crafting [42], where employees engage in proactively re-designing their work to achieve a better fit between their jobs’ demands and resources in an attempt to enhance their work and performance [1, 31]. Structured technologies, such as databases, provide the capability to record, organize, and obtain information. They provide good ways to capture and manage explicit knowledge but not implicit or tacit knowledge that might even have incorrect assumptions [39].

Knowledge acquisition and diffusion. Using GenAI to support knowledge acquisition (by access to an extremely large amount of data or pieces of knowledge) can result in organizational and individual learning. Time savings can be realized with less search efforts (digital or even manual). The organizational value lies in the possibility of reallocating resources within the organization to perform other tasks and functions. Knowledge dissemination and exchange is vital for innovation and competitiveness [10].

Knowledge synthesis. Knowledge may be presented in various formats, including summarization and synthesis leading to further time savings and possibly high levels of accuracy and quality. It may streamline and make available an organization's knowledge resources for its workforce. Competitive efforts can be supported by swift knowledge distribution [38]. Other documented value can be realized, such as enhancing coding knowledge, asking the right questions, and finding additional use cases.

Applications in multiple domains. To additionally assess the value of GenAI, input was solicited from various adopters across a diverse set of companies (marketing, AI start-up), creatives (writer, set designer), and a stockbroker. The respondents were using ChatGPT and were quite consistent in identifying its value as providing a good first draft of either a text document or generated code. Assessment and modification were required, but the time saved to devote to other tasks was substantial. A writer reported that output from GenAI provided a very good initial document, saving a great deal of time, and noted the importance of asking the right questions. Another commented on the considerable effort needed for prompt engineering and the desire to retain their intellectual curiosity.

For document generation, one knowledge worker concluded that GenAI was enormously useful to save time in conducting the fundamental research needed on a topic. However, a significant amount of effort was required to trim the output (too much repetitive information). There were also recognizable, specific words that appear to be overly used, which detracted from the readability of the work. A stockbroker programming specific trades commented on the enormous savings in time to generate code. He suggested that engineers could be relieved from mundane aspects of programming and knowledge workers, in general, eliminated from mundane tasks. This could free up knowledge workers for more interesting problems that require their expertise and critical thinking. A marketing company CEO noted the importance of employees being able to quickly create a marketing campaign for customers, which required human refinement. The latter is an example of a knowledge worker transiting to an expert without progressing through the detailed knowledge acquisition process. However, although there was support for the knowledge acquisition, the worker still had to learn how things are done (process model) at their company and conform to the standards dictated by the company.

3.4 Risks

Both novice and expert knowledge workers are important to an organization. Knowledge workers progress from being novices to experts. Novices rely on the acquisition of explicit knowledge (facts, organizational rules), whereas experts are more skilled, possessing the capacity to synthesize knowledge that might be more based on heuristics or multiple case studies. Expert knowledge workers have tacit or implicit knowledge that is valuable to an organization. As novice knowledge workers adopt GenAI, its use might help them to progress faster to expert knowledge workers. Expert knowledge workers might use GenAI for more complex tasks, but be inclined to assess the validity of the results obtained. It might be possible for novices to use GenAI to progress more quickly to become expert knowledge workers. However, the result could be a loss of the deep organizational memory, which, traditionally, is gained over time by knowledge workers.

Models are increasingly larger and trained on larger amounts of data. This makes it difficult to detect possible biases and/or vulnerabilities. Understanding how AI models work is needed to detect the source of those biases. Developing explainability and interpretability techniques is difficult but critical. LLMs search for patterns, perform a transformation on them, and then apply that source data to produce (generate) some target, required data. Such transformations are often called reasoning but could, perhaps, be more accurately described as guessing, hypothesizing, or abducting. The notion of “understanding” is difficult to correctly interpret for LLMs. It implies the existence of human capabilities that computer systems do not have. LLM might be better described as being able to process, translate, respond, and so on.⁵

Further risks, from an organizational perspective, may be over-reliance on GenAI, re-definition of job requirements and expectations, and modifications to tasks. Technically, large language models are not actually “models of language,” but rather, “statistical models of regularities found in language” [35], implying caution. Of course, these risks and challenges need to be balanced against the access to the vast amount of general and domain-specific knowledge GenAI can provide. There are other known problems. GenAI systems may hallucinate or may provide an incorrect solution. There are always concerns about how to identify or assess the value or possible decrease in value. For example, ChatGPT has documented examples of errors and incorrect or made-up answers, which, without verification, could result in costly or poor decision making.

Knowledge shifting. Expert knowledge workers gain a different type of knowledge than novice workers. They turn tacit knowledge into explicit knowledge, based on their experience. They also gain heuristics and “know-how” for their work and are thus able to apply their common-sense knowledge and intuition as they tackle important problems. GenAI might serve as an internet-wide source of information that has been indexed and organized and readily available for the knowledge worker. Some organizations are creating their own GenAI. This *knowledge shifting* might bypass an important acquisition process. Traditionally, within an organization, novice workers acquire the types of knowledge that lead to their gaining of expertise over time. As noted, when this traditional type of knowledge acquisition goes away (because novice knowledge workers simply retrieve the information they seek from GenAI), the result could be the risk of a loss, over time, of organizational knowledge.

Capabilities of LLMs

There are challenges in the use of LLMs, because they are based on large amounts of data that has been obtained from scraping the internet and attempt to appear to have human-like capabilities. LLMs are challenged by rare words, overfitting, and dealing with complex linguistic rules [8]. Because the data is obtained from crawling the web, there are also concerns regarding copyright, privacy, and bias in the data.

4 Discussion: Impact and Implications

Knowledge management is increasingly important for promoting the creation and sharing of knowledge to support a company [32]. The need to identify what knowledge is needed and how it can be acquired, disseminated, applied, and shared has long been acknowledged [38, p.319]. In the past, this has been difficult because of the limitations of natural language. For decades, most **natural language processing (NLP)** research was based on generative linguistics, psycholinguistics, cognitive linguistics, or logical semantics [34]. However, these were top-down approaches that required starting from some generally agreed-upon principles, with most work based on assumptions about thoughts and language that might not always be true [36]. The notion behind large

⁵This characterization is by John Sowa, via Ontology Form.

Table 1. GenAI Potential Organizational Impact

	Application of GenAI	Task impact	Organizational impact
Use	Summarize audio meetings (virtual meetings)	Provides summary and documentation to participants	Working document of meeting events may be produced quickly with categorization of topics [29]
	Draft reports (legal documents, recommendations, e-mails)	Provides a starting point	Enables knowledge worker to focus on most important content [29]
	Draft of client requirements	Requires less sorting and organizing	For client interaction, speed up time
Value	Micro time savings at scale	—Speeding up task (simple or complex) by a small amount of time can save significant amounts over time	Sum of marginal gains may be large ⁷
	Structured and mundane tasks	—Generate code or summary notes (as above) —Draft cover letters	—Close to full automation of certain tasks might be undesirable to some workers; alternatively, greater retention because employees relieved from such tasks [19] —Streamline tools
	Domain-specific tasks	—Generate drafts of contracts —Generate drafts of client tasks that require creatives (writers, producers)	—Access to large amount of general and domain-specific knowledge upon which to start to address tasks [13]
	Knowledge acquisition	—Learn on demand; enhance pattern recognition capabilities —Gain confidence in knowledge task —Ask for explanations and references —Articulate ideas the GenAI embellishes	—Should be especially helpful for novice knowledge workers —Expand idea generation and obtain knowledge to support decision making [28] —Gain knowledge faster and collaboratively [27]
	Knowledge shifting	—Novice knowledge workers enhance their own knowledge and skills	—Progression from novice to expert proceeds in a different manner and more quickly than in the past
	Financial	—Grow organization without adding more workers	—Impact financials by lowering personnel costs
	Productivity, efficiency, innovation	—Automate tasks	—Reduce costs, provide access to information —Boost productivity by requiring less technical skills [29]

language models is that they are bottom-up reverse engineering of natural language [36]. They can be predictive and generative and are now able to respond to the types of queries that only knowledge workers responded to in the past.

Table 1 summarizes representative applications and impacts of GenAI. As future capabilities of GenAI become more powerful, their adoption and organizational impact will likely continue. Organizations will need to address how to use, manage, and co-create value to benefit well from GenAI.⁶

⁶I thank the anonymous reviewer for this suggestion.

⁷Example from *The Economist*: “I estimate it will save me about a minute per chart. Given that we produce around 60 charts a week, that adds up to around 50 hours a year, or a whole week of time saved.”

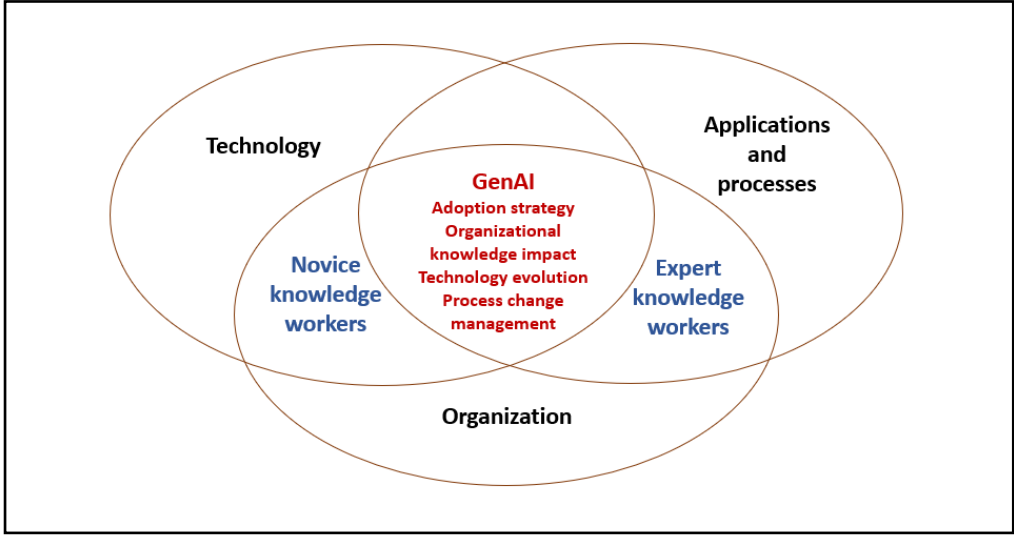


Fig. 2. Overlapping management considerations for GenAI.

Companies are also interested in creating internal large language models. One company is attempting to do so as part of its efforts to use GenAI (ChatGPT), with employees instructed to use GenAI to find opportunities to speed up or automate their jobs [46].

5 Future Research

Businesses the world over are already experimenting with, and adopting, generative AI as they attempt to continue to realize the impact and value of AI tools and techniques.⁸ Knowledge may provide more value to a company than, for example, physical assets [39]. Thus, knowledge management can serve as a firm foundation from which to explore future topics and concepts, since it recognizes the value of knowledge-based assets that can help to create and, equally important, maintain a competitive advantage [2, 12].

As with any potentially disruptive technology, there can be risks taken and value realized. From an organizational perspective, this technology can impact the structure of an organization, how it conducts its business, and its workforce, including its knowledge workers. Figure 2 shows these overlapping considerations.

Table 2 summarizes proposed, needed future research for continued GenAI adoption, broadly based on organizations, technology, and applications.

6 Conclusion

This research has examined knowledge management and the role that generative AI might play in shaping its middle management as knowledge workers progress from novice to expert. The framework for knowledge management and GenAI adoption shows that novice knowledge workers may rely on Generative AI to provide explicit knowledge initially, but then need to progress to gain expertise. Risks include the potential dilution of deep organizational memory due to an over-reliance on GenAI. GenAI will only continue to improve, as will the capabilities they provide, including access to explicit knowledge, support for training, and automation of some or parts of

⁸Of course, major challenges have been identified for generally understanding the risks and impact of AI (e.g., V. C. Storey et al. Explainable AI: Opening the black box or Pandora's Box? *Communications of the ACM* 65, 4 (2022), 27–29).

Table 2. Future Research for Knowledge Management and GenAI

Issues	Research Topics
<i>Organization</i>	
Effort and costs	—How can the cost and benefits of creating a company's own GenAI be assessed? —What parameters and estimates are needed to support such an assessment?
Change management	—How can good, continuous change management that incorporates GenAI be developed within organizations? —Is it possible to make explicit the accomplishments of employees and highlight successful cases?
Skill requirements	—How can GenAI support the progression of skill acquisition from novice to expert? —Will crucial, implicit organizational knowledge get lost when adopting GenAI?
Intellectual property and copyright issues	—How will potential challenges of companies, such as OpenAI, with respect to intellectual property and copyright infringements, impact an organization? What can be done to mitigate such risks?
Impact	—To what extent can GenAI impact overall productivity and innovation within an organization? Will it support training, make knowledge work more enjoyable, or make knowledge workers more confident, leading to increased productivity, both individually and for an organization as a whole? —Does the impact differ by different types of organizations or the industries within which they reside? Is it even possible to generalize by industry or some other coarse level of classification of organizational types, knowledge types, or worker types? —What kinds of knowledge types are needed besides implicit and explicit? —How can the true impact of GenAI be demonstrated to management? What parameters are needed for an assessment? —What are business adoptions, regulators' regulations, or technologists' improvements? What are the unexpected uses, benefits, risks and costs, job effect and potential to impact political situations.⁹
Implications	—If the "secret of business is to know something that nobody else knows" (Aristotle Onassis) what are the implications for competitive advantage if companies have access to the same GenAI technologies?¹⁰ —What are the implications of how to integrate knowledge management with information technologies? —What is the potential for an even bigger Mount Stupid [20] when everyone has access to the same information?¹¹
<i>Applications</i>	
Adoption	—How do we categorize which tasks can be most meaningfully tackled using GenAI? —Will knowledge workers want to (opt-in) use GenAI? —Will knowledge workers be able to assess the value of the support provided by GenAI? —How can top management be trained with respect to: what they should know about GenAI; how they might incorporate it into their workforce; what results might be realized; and the potential downfalls of incorporating this technology, including a loss of valuable organizational knowledge? —What are the well-known risks of GenAI, generally, and how might they impact knowledge workers?

(Continued)

⁹November 6, 2023, Editor, World Ahead, The Economist.

¹⁰"Contributions to sustainable competitive advantage require that resources are valuable, rare, and imperfectly imitable." This should include knowledge. M. Alavi and D. E. Leidner, "Knowledge management and knowledge management systems: Conceptual foundations and research issues." *MIS Quarterly*, 25, 1 (2001), 107–136; J. Barney, "Firm resources and sustained competitive advantage." *Journal of Management*, 17, 1 (1991), 99–120.

¹¹Mount Stupid is the part of the Dunning-Kruger Effect Curve where confidence increases rapidly as one's knowledge of a task or topic moves from novice to amateur before realizing that their ability or knowledge is less than they thought (valley of despair) and then progressing to enlightenment.

Table 2. Continued

Issues	Research Topics
<i>Applications</i>	
Knowledge	—Will notions of the types of knowledge need to be redefined? Besides implicit (tacit), explicit, and structural, are other classifications needed? —What exactly is “GenAI knowledge” and how should it be defined, if it actually differs from known knowledge types?
<i>Technology</i>	
Expectations	—Will GenAI live up to its, possibly over-inflated, expectations? ¹²
Hallucinations	—How can knowledge workers identify potential hallucinations from GenAI, given the nature of the data and algorithms? —How can the impact of hallucinations be identified and assessed [23, 44]? —Is it possible to avoid doing things that could have a negative impact on an organization because of hallucinations that are undetected? —How can knowledge workers be trained to identify and verify before adoption/adaption of the output from GenAI?
Data-driven, bottom-up, non-deterministic	—How will an appreciation of the bottom-up and non-deterministic nature of GenAI [34] affect potential knowledge work use?

knowledge workers’ tasks. The implication of GenAI use and value will also evolve. Additional research is needed to further understand the potential impacts of GenAI adoption with respect to the knowledge workers, the organization, and the use of technology.

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¹²<https://www.ibm.com/think/insights/gen-ai-live-up-to-hype>

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